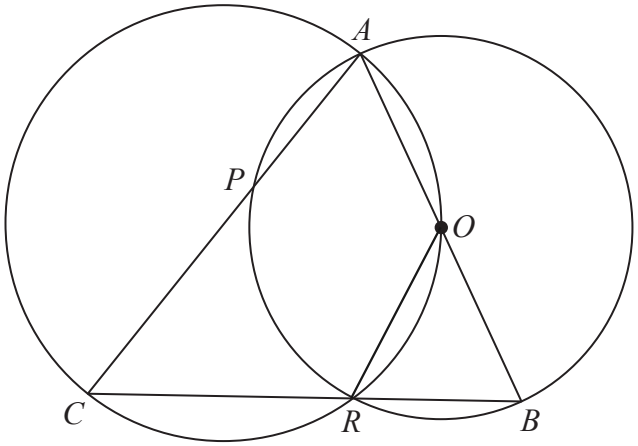
21



A, B, R and P are four points on a circle with centre O.
A, O, R and C are four points on a different circle.
The two circles intersect at the points A and R.

CPA, CRB and AOB are straight lines.

Prove that angle CAB = angle ABC.

Let angle CAB = x. Since A, O, B are collinear, CAO = x.

Angle CRO = 180° – x

Opposite angles of cyclic quadrilateral CAOR add to 180°.

1 mark
Use the cyclic quadrilateral

Angle ORB = 180° – (180° – x) = x

Angles on the straight line CRB add to 180°.

1 mark
Use the straight line

OR = OB (radii of the circle with centre O).

So angle OBR = angle ORB = x,

because base angles of an isosceles triangle are equal.

1 mark
Use equal radii and base angles

ABC = OBR = x = CAB, since BA and BO are the same ray and BC and BR are the same ray. Therefore CAB = ABC.

1 mark
Complete the proof with all reasons linked

(Total for Question 21 is 4 marks)